Defining Clinically Interpretable Glycemic Profiles Using Continuous Glucose Monitoring in Type 2 Diabetes

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The American Diabetes Association (ADA) recommends continuous glucose monitoring (CGM) metrics for monitoring glucose control, including time in range (TIR) (70–180 mg/dL), time below range (TBR) (<70 and <54 mg/dL), time above range (TAR) (>180 and >250 mg/dL), mean glucose, and glucose coefficient of variation (CV) (1). Cluster analysis of these clinical CGM metrics may help us uncover distinct glycemic subgroups.

The high volume and temporal resolution make CGM data well-suited for data-driven approaches (e.g., clustering) to uncover latent CGM profiles. Prior clustering studies have used CGM metrics with limited clinical interpretability (e.g., J-index) (2,3). ADA-recommended CGM metrics are more clinically intuitive but may not capture the full richness of CGM data. This highlights an opportunity to find a middle ground between complex, research-oriented CGM metrics and ADA-recommended clinical metrics for clustering.

We aimed to identify distinct CGM profiles through clustering on a focused set of CGM metrics used in clinical practice, supplemented by additional interpretable percent time (e.g., TIR 70–140 mg/dL) and diurnal metrics (e.g., day/night TBR <70 mg/dL), in people with type 2 diabetes (T2D) not treated with insulin from the Hyperglycemic Profiles in Obstructive Sleep Apnea (HYPNOS) trial (4). We externally validated these profiles in the Atherosclerosis Risk in Communities Generation 2 (ARIC Gen2) study (5). We also examined whether the identified CGM profiles provided information on glycemic control beyond hemoglobin A1c (HbA_{1c}).

We derived 27 CGM features from up to 14 days of CGM data (Dexcom G4 Platinum) in HYPNOS participants, including TIR 70–180 mg/dL, TBR <70 and <54 mg/dL, TAR >180 and >250 mg/dL, sensor mean glucose, and CV (1). We included additional percent time metrics and SD to capture diurnal glycemic patterns and more detailed variability (Supplementary Methods).

We applied an iterative robust and sparse k-means clustering to identify CGM profiles (Supplementary Methods). External validation was assessed using an adjusted Rand index, which measures agreement between clustering results (values closer to 1 indicate higher agreement). We examined demographics, clinical characteristics, and diabetes medication use. All analyses were performed in R 4.2.2.

From the 182 participants (mean age of 60 years, 51% female, and 34% Black adults), we identified five CGM profiles. Based on ADA-recommended CGM metrics, we named clusters as follows: "hypoglycemia high-variability," "moderate-hypoglycemia moderate-variability," "well-controlled," "moderate-hyperglycemia moderate-variability," and "hyperglycemia low-variability" (Fig. 1).

The smallest cluster, hypoglycemia high-variability, had mean values of TBR <70 mg/dL, TBR <54 mg/dL, and CV exceeding ADA targets: 11.5%, 4.1%, and 37.9%, respectively (Supplementary Table 1). The moderate-hypoglycemia moderate-variability group had TBR and CV within targets but approaching upper limits, while the well-controlled group met all ADA targets. Despite having similar median HbA_{1c} (∼7%), these three clusters exhibited markedly different CGM patterns, ranging from frequent hypoglycemia and high variability to a stable and well-controlled glucose profile. The hyperglycemia low-variability group had the highest TAR and HbA_{1c} levels, followed by the moderate-hyperglycemia moderate-variability group (Fig. 1).

The hyperglycemia low-variability group had the lowest average age (Supplementary Table 2). The hyperglycemia low-variability group had the highest levels of liver enzymes and triglycerides. The hypoglycemia high-variability group had the highest HDL cholesterol. The moderate-hypoglycemia moderate-variability group and the well-controlled group had the lowest estimated insulin resistance, whereas the hyperglycemia low-variability group showed

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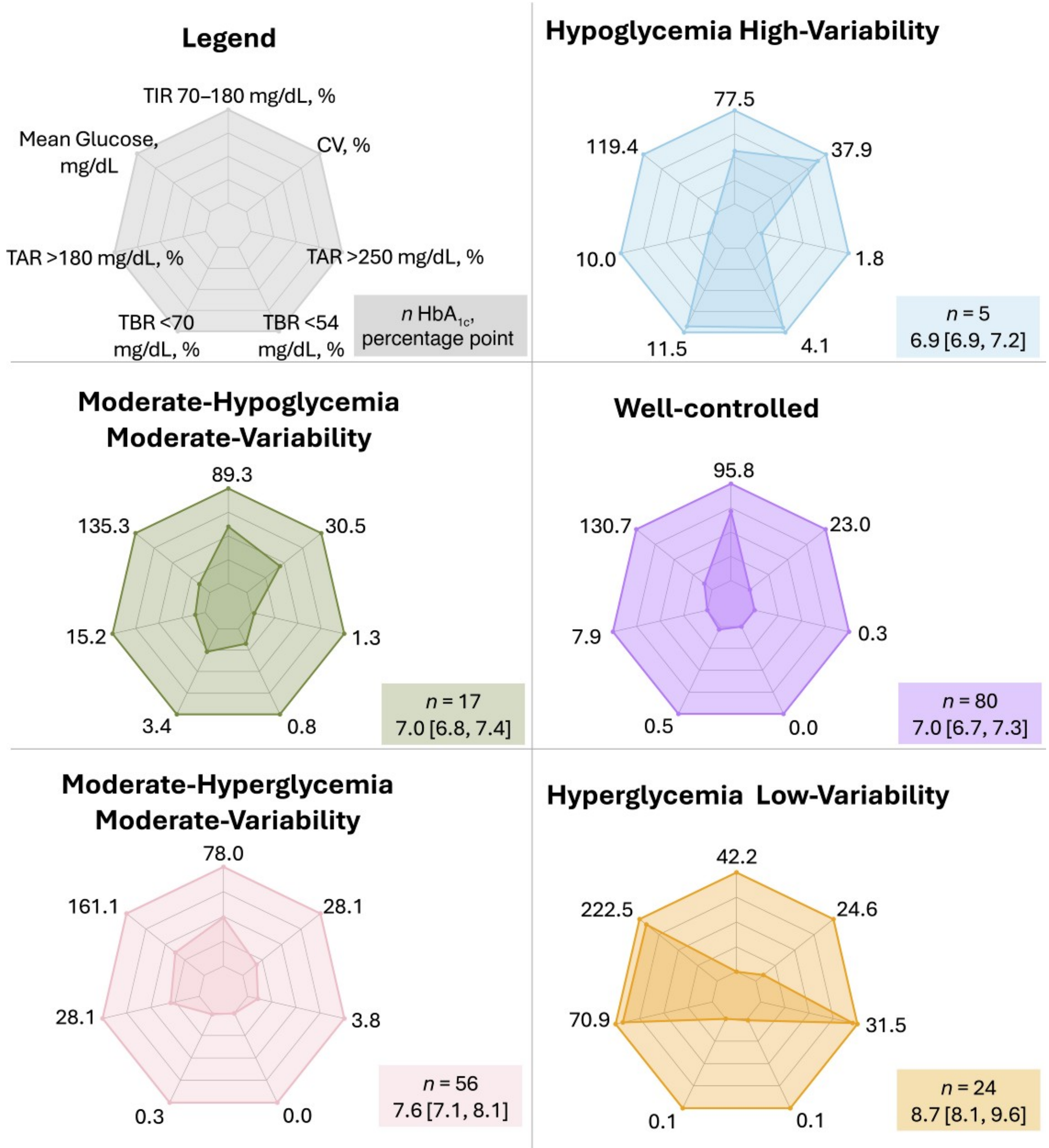

Figure 1—Visual displays of glycemic profiles in HYPNOS in radar plots and HbA_{1c}. CGM metrics are presented as means; HbA_{1c} is presented in median [interquartile interval]. Mean glucose is in mg/dL. TIR, TBR, and TAR are all in percent of total CGM wear time. CV is in percent. HbA_{1c} is in percentage point. TAR, time above range.

the highest levels of insulin resistance and lowest estimated β-cell function. Sulfonylurea use was the highest in hypoglycemia high-variability group (80%) and lowest in the well-controlled group (15%).

External validation in ARIC Gen2 showed an adjusted Rand index of 0.56, suggesting moderate reproducibility of the CGM clusters. This modest agreement may partly reflect differences in cohort characteristics, as ARIC Gen2 participants were older (mean 66 years), included more women (61%), had lower median HbA_{1c} (6.8%), and lower use of sulfonylureas (17%) or metformin (54%) (Supplementary Table 3).

Previous CGM clustering studies identified diabetes subtypes using complex CGM metrics with limited clinical interpretability (2,3). We extended this work by showing that clustering with a clinically intuitive feature set could identify five distinct CGM profiles, which we labeled with ADA-recommended metrics, aiming to further enhance the interpretability and potential clinical utility of CGM clustering. We performed external validation and demonstrated moderate reproducibility. Limitations of the study include the small sample size, leading to potentially uneven cluster distribution, and lack of clinical outcomes.

In conclusion, using clustering with a set of clinically interpretable CGM metrics, we identified five distinct CGM profiles in adults with T2D not treated with insulin that provided information on glucose patterns not reflected in HbA_{1c} and had differing clinical characteristics (e.g., insulin resistance, β-cell function, lipid levels, and liver enzymes). Our results suggest the value of clinical CGM metrics in data-driven glycemic profiling, which may inform personalized diabetes management strategies.

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